

Tascs:

1. Add some records in the existing table
2. Apply window function on the data
3. Apply View on the data
4. Apply Triggers on the data

Aswer:

Here we have only two tables first is

- 1.employee
- 2.department

Let's see the tasc

1. Add some records in the existing table

SELECT * FROM sagar.department;

```
insert into sagar.department
(department_id,department_name,location)
values(8,'finance','nagpur'),
(9,'worker','mumpuri'),
(10,'public speaker','pune');
```

SELECT * FROM sagar.department;

Result Grid	Filter Rows:	Edit:
department_id	department_name	location
1	sells	latur
2	IT	pune
3	finance	nanded
4	HR	panjab
5	other	Beed
6	marketing	latur
7	clening staff	solapur
8	finance	nagpur
9	worker	mumpuri
10	public speaker	pune
NULL	NULL	NULL

department 8 x

select * from sagar.employee;

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

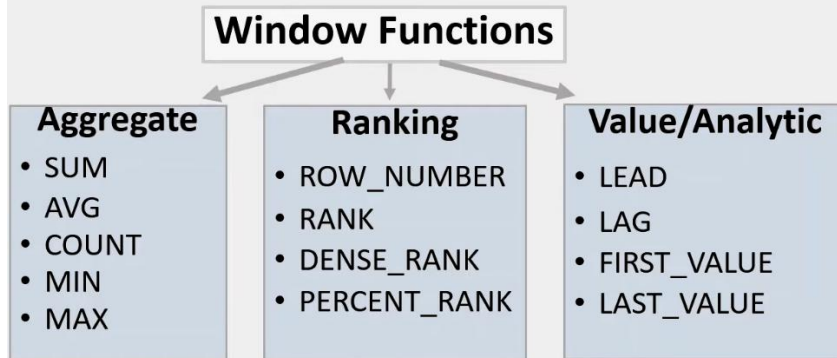
	emp_id	name	department	salary	hire_date
▶	1	sagar dixit	IT	100000	2020-05-25
	2	Sham sawant	sells	10000	2020-04-02
	3	Ram bhosle	HR	50000	2022-04-12
	4	kartik Aryan	IT	40000	2022-01-09
	5	Raghav wag	sells	67000	2022-01-11
	6	Raj mene	IT	35000	2024-05-30
	8	smith	marketing	30000	2026-01-20

2. Apply window function on the data

There are three types of the windows function

WINDOW FUNCTION TYPES

There is no official division of the SQL window functions into categories but high level we can divide into three types



Here we use 1 aggregate function

```
select department,salery ,sum(salery)  
over (partition by department order by emp_id )  
as total_salery from  
sagar.employee;
```

Result Grid			
Filter Rows:			
department	salary	total_salary	
HR	50000	50000	
IT	100000	100000	
IT	40000	140000	
IT	35000	175000	
marketing	30000	30000	
sells	10000	10000	
sells	67000	77000	

select department,salary ,

AVG(salary) over (partition by department order by salary)
AS 'AVRAGE',

SUM(salary) over (partition by department order by salary)
AS 'TOTAL',

MIN(salary) over (partition by department order by salary)
AS 'MINIMUM_SAL',

MAX(salary) over (partition by department order by salary)
AS 'MAX_SAL',

COUNT(salary) over (partition by department order by
salary) AS 'COUNT'

FROM sagar.employee;

```

7 • select department,salary ,
8     AVG(salary) over (partition by department order by salary) AS 'AVRAGE',
9     SUM(salary) over (partition by department order by salary) AS 'TOTAL',
10    MIN(salary) over (partition by department order by salary) AS 'MINIMUM_SAL',
11    MAX(salary) over (partition by department order by salary) AS 'MAX_SAL',
12    COUNT(salary) over (partition by department order by salary) AS 'COUNT'
13
14    FROM sagar.employee;

```

	department	salary	AVRAGE	TOTAL	MINIMUM_SAL	MAX_SAL	COUNT
▶	HR	50000	50000.0000	50000	50000	50000	1
	IT	35000	35000.0000	35000	35000	35000	1
	IT	40000	37500.0000	75000	35000	40000	2
	IT	100000	58333.3333	175000	35000	100000	3
	marketing	30000	30000.0000	30000	30000	30000	1
	sells	10000	10000.0000	10000	10000	10000	1
	sells	67000	38500.0000	77000	10000	67000	2

Here we use 1 RANKING

select salary ,

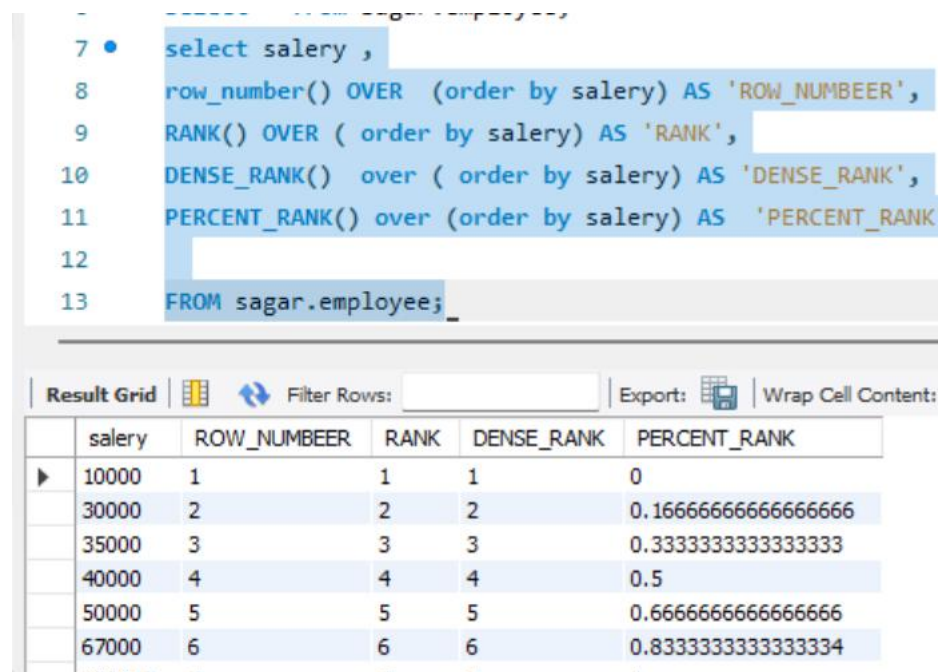
row_number() OVER (order by salary) AS
'ROW_NUMBEER',

RANK() OVER (order by salary) AS 'RANK',

DENSE_RANK() over (order by salary) AS
'DENSE_RANK',

PERCENT_RANK() over (order by salary) AS
'PERCENT_RANK'

FROM sagar.employee;



```
7 • select salary ,
8     row_number() OVER (order by salary) AS 'ROW_NUMBEER',
9     RANK() OVER ( order by salary) AS 'RANK',
10    DENSE_RANK() over ( order by salary) AS 'DENSE_RANK',
11    PERCENT_RANK() over (order by salary) AS 'PERCENT_RANK'
12
13 FROM sagar.employee;
```

	salary	ROW_NUMBEER	RANK	DENSE_RANK	PERCENT_RANK
▶	10000	1	1	1	0
	30000	2	2	2	0.16666666666666666
	35000	3	3	3	0.3333333333333333
	40000	4	4	4	0.5
	50000	5	5	5	0.6666666666666666
	67000	6	6	6	0.8333333333333334

Here we use 1 ANALYATICAL function

select salary ,

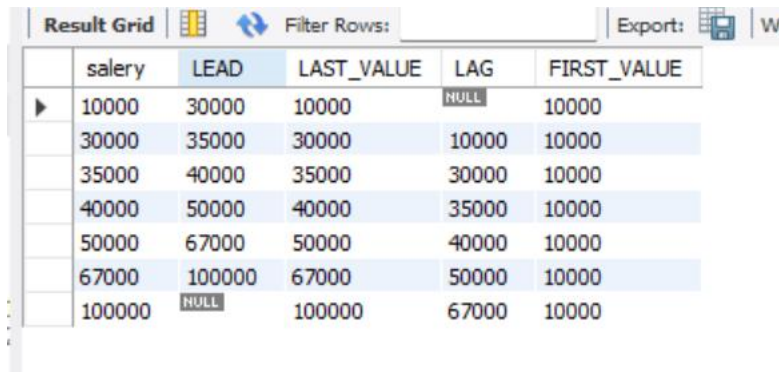
lead(SALERY) OVER (order by salary) AS 'LEAD',

last_value(SALERY) over (order by salary) AS
'LAST_VALUE',

LAG(SALERY) over (order by salary) AS 'LAG',

FIRST_VALUE (SALARY) over (order by salary) AS
'FIRST_VALUE'

FROM sagar.employee;



The screenshot shows a 'Result Grid' window with a toolbar at the top containing 'Filter Rows:', 'Export:', and a 'W' icon. The grid displays the following data:

	salary	LEAD	LAST_VALUE	LAG	FIRST_VALUE
▶	10000	30000	10000	NULL	10000
	30000	35000	30000	10000	10000
	35000	40000	35000	30000	10000
	40000	50000	40000	35000	10000
	50000	67000	50000	40000	10000
	67000	100000	67000	50000	10000
	100000	NULL	100000	67000	10000

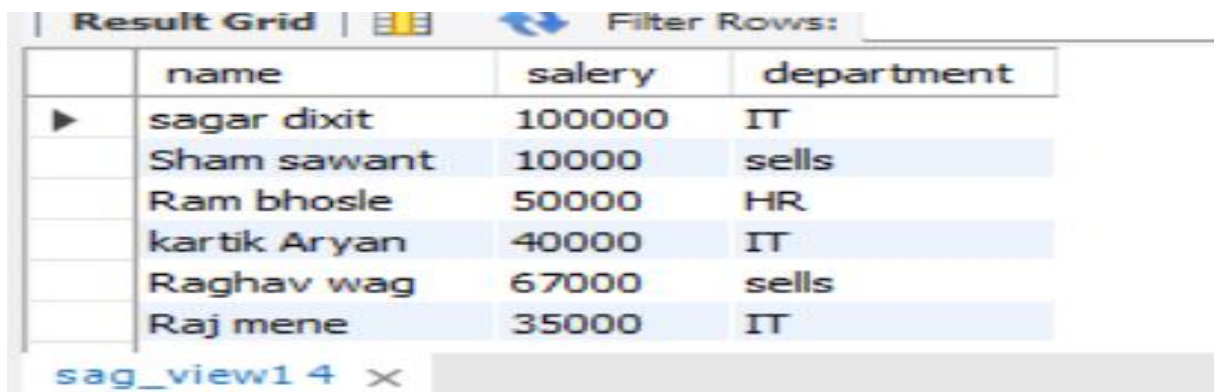
3. Apply View on the data

create or replace view sag_view1

as select name ,salery,department

from sagar.employee;

select * from sag_view1;



The screenshot shows a 'Result Grid' window with a toolbar at the top containing 'Filter Rows:'. The grid displays the following data:

	name	salery	department
▶	sagar dicit	100000	IT
	Sham sawant	10000	sells
	Ram bhosle	50000	HR
	kartik Aryan	40000	IT
	Raghav wag	67000	sells
	Raj mene	35000	IT

sag_view1 4 x

drop view sag_view1;

```
drop view sag_view1;
```

n Output

Time	Action	Message
08:48:38	select * from sag_view LIMIT 0, 1000	3 row(s) returned
08:54:08	create or replace view sag_view as select name ,salary,department from sager.emp...	Error Code: 1049. Unknown
08:54:29	create or replace view sag_view1 as select name ,salary,department from sager.em...	Error Code: 1049. Unknown
08:54:51	create or replace view sag_view1 as select name ,salary,department from sagar.em...	0 row(s) affected
08:55:38	select * from sag_view1 LIMIT 0, 1000	7 row(s) returned
09:01:51	drop view sag_view1	0 row(s) affected

4. Apply Triggers on the data

DELIMITER //

CREATE TRIGGER trigger_after_insertion

BEFORE INSERT ON sagar.employee

FOR EACH ROW

BEGIN

IF NEW.salary < 0 THEN

SET NEW.salary = 0;

END IF;

END;

//

DELIMITER ;

INSERT INTO sagar.employee (emp_id, name, department,
salary, hire_date)

VALUES (9, 'ram', 'IT', -80000, '2001-04-03');

select * from sagar.employee;

Result Grid |   Filter Rows: | Export:  | Wrap

emp_id	name	department	salary	hire_date
1	sagar dixit	IT	100000	2020-05-25
2	Sham sawant	sales	10000	2020-04-02
3	Ram bhosle	HR	50000	2022-04-12
4	kartik Aryan	IT	40000	2022-01-09
5	Raghav wag	sales	67000	2022-01-11
6	Raj mene	IT	35000	2024-05-30
8	smith	marketing	30000	2026-01-20
9	ram	IT	0	2001-04-03

ployee 2 ×